SML Quiz 4

Total Marks: 10
Duration: 60 minutes

Name: ______________________________ **Roll No.:** ____________________

Course Outcomes

CO1	Students will be able to understand the various key paradigms for machine learning and pattern classification
CO2	Students will be able to apply suitable feature extraction and classification technique to solve a given pattern classification problem
CO3	Students will be able to design a complete machine learning/pattern classification algorithm and evaluate the performance

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Q1. A parent node contains 10 samples: [CO3] [2.5]

6 positives, 4 negatives

Two candidate splits are proposed.

Split A:

- Left child: 4+, 1−
- Right child: 2+, 3−

Split B:

- Left child: 5+, 3−
- Right child: 1+, 1−

Using the Gini index,

(a) compute the weighted Gini index after Split A,
(b) compute the weighted Gini index after Split B,
(c) state which split is better.

Q2. A bagging classifier is trained using 5 trees on four training points x_1, x_2, x_3, x_4. The OOB predictions are given below. [CO3] [2.5]

Tree	**OOB points and predictions**
T_1	$x_1 \to +,\ x_3 \to -$
T_2	$x_2 \to -,\ x_3 \to +,\ x_4 \to -$
T_3	$x_1 \to +,\ x_4 \to +$
T_4	$x_2 \to -,\ x_3 \to -$
T_5	$x_1 \to -,\ x_2 \to +,\ x_4 \to -$

The true labels are

$$x_1 : +, \qquad x_2 : -, \qquad x_3 : -, \qquad x_4 : +$$

(a) For each point, obtain its OOB majority vote.

(b) Compute the OOB error.

Q3. Consider 5 training examples with labels [CO3] [2.5]

$$y = [+1, +1, -1, -1, +1]$$

Initially, all weights are equal:

$$w_i^{(1)} = \frac{1}{5}, \qquad i = 1, \ldots, 5$$

In round 1, a weak learner predicts

$$h_1(x) = [+1, -1, -1, +1, +1]$$

(a) Compute the weighted error

(b) Compute α_1

Now in round 2, suppose a second weak learner predicts

$$h_2(x) = [+1, +1, -1, -1, -1]$$

(c) Compute the weighted error for round 2.

Q4. Consider MSE error in place of 0-1 error in AdaBoost. Give an expression for weighted MSE loss. Derive α in terms of predictor at round j, weights w_i, labels y_i, and predictor score till $(j-1)^{th}$. Assume the weights to be normalized to add to 1, so that the denominator term vanishes. [CO1][2.5]